

A Simple Rust Model

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1 Introduction

In this document we develop a simple Rust model so we can show the power of JAX when using a NFXP algorithm for estimation.

2 The Model

The model follows closely the version from Hotz (1994). In each month t , the owner decides whether or not to replace a bus's engine ($J = 2$) to minimize the discounted costs of maintaining it. Everytime he is faced with a bus, he observes its accumulated mileage x_t and decides the replace decision, $d_t = 1$ if true and $d_t = 0$ otherwise.

The costs are as follows:

$$u_{t1} = -\beta^t(\theta_0 + \theta_1 x_t + \varepsilon_{t1})$$

$$u_{t2} = -\beta^t(\theta_1 x_t + \varepsilon_{t2})$$

Where ε_t is an iid Type I Extreme value random variable. The law of motion governing mileage is:

$$x_{t+1} = b_t + (1 - d_t)x_t$$

Where b_t is an iid random variable with discrete support $\{0, 1, 2\}$. For our commodity, we assign fixed probabilities for each scenario that do not depend on x_t . The probabilities are 0.1, 0.5 and 0.4 repectively. The mileage has an upper limit, such that $x_t \in \{1, 2, \dots, 90\}$. When in mileage $x_t = 89$ or $x_t = 90$, the probabilities group together until the probability to increase above 90 is zero.

The bus manager sequentially chooses for each bus $\{d_{t1}\}_{t=0}^{\infty}$ such that he maximizes

$$E_0 \left[\sum_{t=0}^{\infty} -\beta^t [d_t(\theta_0 + \theta_1 x_t + \varepsilon_{t1}) + (1 - d_t)(\theta_1 x_t + \varepsilon_{t2})] \right]$$

The problem can be expressed in a bellman equation with x_t as the state variable.

$$V(x_t) = \max_{d_t} [v_{td} + \varepsilon_{td} + \beta E_t V(x_{t+1})]$$

Where v_{td} is the choice specific value function. Let's inspect the payoffs of each choice so we are clear what we are modelling here:

$$V(x_t) = \begin{cases} \theta_0 + \theta_1 x_t + \varepsilon_{t1} + \beta(q_0 V(0) + q_1 V(1) + q_2 V(2)) & \text{Replace} = 1 \\ \theta_1 x_t + \varepsilon_{t2} + \beta(q_0 V(x_t) + q_1 V(x_t + 1) + q_2 V(x_t + 2)) & \text{Keep} = 1 \end{cases}$$

Where q_a is the probability of mileage increasing a . This operation is the contraction mapping T of V . Let's rename each entry as v_r , the discounted value of replacing, and v_k , the discounted value of keeping. As it is known, the fixed point of T , V^* , is the solution to the bellman equation. Rust(1987) made this manageable working with EV , the integration over all the stochastic paths, instead of V . Applying the expectation over V and expressing it in log-sum form as our errors are type I extreme value, we end up with the following expression:

$$EV(x_t) = \log(e^{v_r(EV)} + e^{v_k(EV)})$$

Once we find the fixed point EV^* we can evaluate $v_r(EV)$ and $v_k(EV)$. We then obtain the probabilities which have a logit form. For simplicity we use the formula:

$$Pr(d = 1) = \exp(v_r - EV)$$

Proof follows from the logit specification of probability and our log-sum equation.

With all of the above we get the probabilities from the primitives.

3 Data Generation

4 Monte Carlo

5 Contact

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